Ishaq Shaik

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Experience

Ford Motor Company Sep. 2026 – Present
Software Engineer - ML *San Francisco, CA*

- Built model performance dashboards in **Grafana** for Power-Up software updates, tracking accuracy drift across 1M+ vehicles.
- Developed active learning filters for BlueCruise camera data, prioritizing edge cases to reduce annotation costs by 25%.

Arizona State University May 2025 – May 2026
Graduate Research Assistant *Tempe, AZ*

- Deployed a RAG chatbot on **AWS Lambda** serving **5,000+** users at **1.2s** latency with schema-validated request handling.
- Cut multi-turn inference cost by **40%** through context summarization, payload packing and secure session state management.
- Owned the chatbot and ML pipeline end to end as the team's sole developer, from design through production deployment.
- Scaled an **AWS Lambda + RDS** ML pipeline to **300k+** events at **99.9%** uptime, automating ingestion from monthly to daily.

Extreme Networks Jul. 2023 – Jun. 2024
Software Engineer *Bangalore, India*

- Achieved **99.99%** state recovery across **100K+** AP devices by architecting a **Java Kafka** pipeline for distributed offline syncs.
- Reduced inter-service latency by **25%** across **1M+** daily requests by migrating Java REST APIs to **gRPC + Protobuf**.
- Designed **Spring Boot** APIs enabling global remote configuration for AP5010 and AP410C access points across client sites.
- Delivered **8** Java Spring Boot data APIs for **20+** NFL/MLB stadium sites, validating 70+ edge cases on flagship hardware.

Nbyula Mar. 2023 – May 2023
Software Engineer Intern *Bangalore, India*

- Cut image load times by **84%** (2.5s to 400ms) by implementing **CloudFront** edge caching and server-side responsive cropping.
- Raised chat DAU by **32%** (3.2K to 4.2K) by fixing **Redis** cache invalidation bugs and optimizing React rendering.

LoadShare Networks Sep. 2022 – Dec. 2022
Software Developer Intern *Bangalore, India*

- Eliminated manual SQL ops saving 15 min/day by building a **Java Spring Boot** Slack bot with automated query execution.
- Accelerated deployment by **75%** (40 to 10 min) by Dockerizing Java Spring Boot microservices with **CI/CD** automation.

Projects

Chess Transformer Lichess Bot | *PyTorch, ONNX, AWS EC2, SLURM • Independent Project* | sonuishaq67/chess | lichess/Q_K_V

- Deployed live on Lichess as bot **Q_K_V**, reaching about **2500** rapid over **5,800+** games served through the Lichess Bot API.
- Trained a **500M**-param decoder-only transformer on **10B** tokens using PyTorch mixed precision on **8x Gaudi2** with SLURM.
- Built an end-to-end Python data pipeline over **100M+** games using DuckDB filtering, UCI tokenization and numpy memmaps.
- Exported the model to **ONNX** for sub-second move inference on AWS EC2, serving real-time games via the Lichess Bot API.

Pong Double DQN Reinforcement Learning | *PyTorch, Gymnasium, CNN • Independent Project* | sonuishaq67/pong

- Achieved **100%** win rate across **10K** games by training a PyTorch **Double DQN** with experience replay and soft target updates.
- Accelerated training by **91%** by parallelizing rollouts across 16 AsyncVectorEnv workers on a GPU cluster with torch.compile.

Amazon Reviews RAG System | *Python, PostgreSQL, pgvector, Neo4j • Class Project*

- Boosted precision@10 by **27%** over **571M** Amazon reviews using a hybrid vector + knowledge graph retrieval pipeline.
- Identified a **20%** precision gain from hybrid retrieval via ablation studies against vector-only and graph-only baselines.

Agent-Debate: Multi-Agent Research System | *Python, OpenAI Agents, ElevenLabs, Beliefs SDK • Hackathon* | agent-debate

- Built a multi-agent orchestrator chaining **5** agents (plan, retrieve, summarize, critic, write) for grounded debate reports.
- Synthesized debate podcasts via **ElevenLabs** TTS, using Beliefs SDK to track per-turn claim confidence across rounds.

Technical Skills

Languages Python, C++, Java, SQL, Go, JavaScript, TypeScript, Bash
ML & AI PyTorch, Transformers, ONNX, Mixed Precision, Reinforcement Learning, RAG, pgvector
Data & Backend PostgreSQL, Redis, Kafka, DuckDB, Neo4j, Spring Boot, FastAPI, gRPC, Protobuf
Cloud & Infrastructure AWS (EC2, RDS, S3, Lambda, SQS), Docker, Kubernetes, SLURM, Linux, Terraform, CI/CD

Education

Arizona State University Tempe, AZ
*Master of Science in Computer Science • GPA: **3.8/4.00*** *Aug. 2024 – May 2026*

Ramaiah Institute of Technology Bangalore, India
Bachelor of Engineering in Information Science *Aug. 2019 – May 2023*